MIL-T-29015(TD)
3 September 1970
SUPERSEDING
NAVTRADEVCEN 3313-427A
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MILITARY SPECIFICATION

# TRAINING AID; MOCKUP, SAFETY VALVE, HUDDLING CHAMBER

This specification has been approved by the Naval Training Device Center, Department of the Navy.

## 1. SCOPE

1.1 This specification establishes the requirements for a three dimensional mockup of a huddling chamber type safety valve, (see 6.1).

1.1.1 This item has been designated by the Naval Training Device Center, as Device 29CC3 for Navy procurement (see 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated-plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-corner |
| PPP-B-636 | Box, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation; Methods of |
| MIL-A-8625 | Anodic Coatings, For Aluminum and Aluminum Alloys |
| MIL-L-10547 | Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible |

FSC 6910

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|---|---|
| MIL-T-23991 | Training Devices, Military; General Specification for |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
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| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-143 | Specifications and Standards; Order of Precedence for the Selection of |

PHOTOGRAPHS

Naval Training Device Center (NAVTRADEVCEN)

| | |
|---|---|
| NP-40-867-8-55, NP-40-868-8-55 and NP-40-3378-4 | Device 29CC3, Huddling Chamber, Safety Valve |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be as specified in 3.1.1 through 3.1.6. Materials not specified herein shall conform to the requirements of MIL-T-23991.

3.1.1 Metals.- Metal parts shall be of a corrosion-resistant metal or of a metal given a corrosion-resistant treatment or coating.

3.1.2 Wood.- Wood used shall be of select grades of hardwood lumber, suitable for either clear or opaque finishes. Wood shall be seasoned, free from checks, splits and crossgrain or seasoning defects which may cause warping.

3.1.3 Glue.- Glue used shall not be adversely affected by moisture, lacquer, paint, or migratory solvents of plastics used in components, including the dust cover of 3.6.5.

3.1.4 Plastics.- Plastics shall be resistant to cracking, brittleness, discoloration, deformation and similar defects, upon aging or when subjected to environmental conditions specified in 3.3.



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3.1.4.1 Plastic cement.- Contact surfaces of plastic pieces to be joined shall be machined or sanded. Wherever possible, a cushion surface shall be developed on one contact surface. Cement solvents shall be used as recommended by the manufacturer of the particular plastic used and tested by the contractor for each specific application to conform to the requirements of 3.3. Cemented joints shall be free of bubbles and form a strong homogeneous bond. Capillary joints shall require prior written approval of the procuring activity.

3.1.5 Plywood.- Plywood shall have a minimum of five plies, or solid core hardwood, both surfaces to be of hardwood, all plies to be free of spaces or checks and joined by moisture resistant glue.

3.1.6 Written approval of the Contracting Officer shall be required for the selection of materials and parts from Standards and Specifications of MIL-STD-143, Groups II, III and IV in accordance with nonstandard materials and parts requirements of MIL-T-23991.

3.2 Design.- The training aid shall be a three-dimensional mockup of a huddling chamber type safety valve, mounted on a panel. The training aid shall be designed as indicated on NAVTRADEVCEN Photographs NP-40-3378-4, NP-40-867-8-55, and NP-40-868-8-55. It shall demonstrate the principles of operation, component arrangement, and identify the major components of a huddling chamber type safety valve. The panel shall have two controls and two scales. These controls shall demonstrate the operation of the huddling chamber safety valve through manipulation of the various components.

3.3 Reliability.- The training aid shall be designed for a useful classroom life of 20 hours per week for 5 years. Reliability shall be a prime consideration in the design and manufacture of the training aid. The design shall be such as to result in reduced requirements for adjustments or alinement. The training aid shall be constructed so no fixed part shall become free and no moving part shall become loose or sluggish in operation. Degradation shall not be caused because of shipment, normal handling in classroom use, or storage. The training aid shall be designed for storage and transportation when under ambient conditions of temperature ranging from -5° to +125° Fahrenheit, relative humidity ranging up to 95 percent, and still be capable of meeting the specification requirements.

3.4 Construction.- Construction of the training aid shall be as specified in 3.5 through 3.7.2. Details of construction not specified shall be in accordance with the processes requirements of MIL-T-23991.

3.5 Dimensions and tolerances.-

3.5.1 Dimensions.- The dimensions shall be consistent with the functional design of the device and inherent qualities of the material being used.

3.5.2 Tolerances.- Tolerances in construction, shall be consistent with functional design, inherent qualities of the material being used, and shall



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afford freedom from binding in structural and mechanical parts that are required to be movable.

3.6 Details of components.- The training aid shall consist of the following major components:

(a) Panel and stand

(b) Mockup

(c) Controls

(d) Nomenclature plate

(e) Dust cover.

Details of the components for the huddling chamber safety valve training aid shall be as shown on the photographs of 3.2 and as specified in 3.6.1 through 3.6.4.

3.6.1 Panel and stand.- The panel and stand shall be constructed from 3/4 inch plywood. The stand shall provide a 7° rearward tilt to the panel. The panel shall be framed on all four sides with aluminum or corrosion-resistant steel edging having a satin finish.

3.6.2 Mockup.- The mockup of the huddling chamber safety valve shall be attached to the front of the panel and shall be a three-dimensional mockup, cross sectioned to show the interior components of the valve in three dimensional form.

3.6.3 Controls.- The training aid shall have two controls and two scales mounted on the front of the panel to demonstrate the operation of the valve. The left-hand control and scale shall provide representative settings for the steam pressure, and the right-hand control and scale shall provide representative adjustments made to the adjusting ring. These two controls shall operate the valve and adjusting ring in a manner which demonstrates the operation of the valve. The two controls shall be linked together in such a manner that the valve opens only when the steam pressure control is moved to a higher reading on its scale than that on the adjusting ring control and scale.

3.6.4 Nomenclature plate.- A nomenclature plate shall be provided which shall be keyed to numbered discs mounted on the valve mockup. (See 3.8.2.1).

3.6.5 Dust cover.- A dust cover equal to the tensile, tear and flame resistance characteristics of Armor-Tite nylon reinforced vinyl, or equivalent material, 16 ounce weight, light gray, shall be provided to cover the training aid.

3.6.6 The dust cover shall be labeled with black lettering of 3/4 inch height to identify the device by title and number.



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3.7 Finish and color.-

3.7.1 Finish.- Surfaces to be painted shall be properly sealed, primed, and then finished with a satin surface lacquer in accordance with the Preparation for Painting and Painting requirement, of MIL-T-23991. Application of finishes shall be accomplished in such a manner as not to impair performance of the training aid.

3.7.1.1 Aluminum parts shall be anodized in accordance with MIL-A-8625.

3.7.2 Colors.- The training aid components, interconnecting parts and fittings, shall be color coded to distinguish steam, structural and mechanical functions, as indicated on the photographs of 3.2. Colors shall be in accordance with FED-STD-595. Color coding shall be as follows:

(a) Edges of cutaway parts shall be primed and painted White

(b) The sides of all parts (unless otherwise shown) shall be primed and painted Gray color #26314

(c) The stand and panel board shall be primed and painted Gray color #26152

(d) The spring and spindle shall be painted Aluminum #17178

(e) The padlock, lever, release nut, top spring perch, bottom spring perch, compression screw, top ring, adjusting ring, and ring pin shall be painted Gray #26492

(f) The feather shall be painted Gray color #26152

(g) The arrows in the steam inlet shall be painted Red color #21105

(h) The arrows on the feather shall be White

(i) The steam inlet area shall be painted Orange color #22246

(j) The steam outlet chamber area shall be painted Orange color #22356.

3.8 Nameplates and product markings.-

3.8.1 Nameplates and emblem.- A Government-furnished training device nameplate and emblem, shall be attached to each training aid as indicated on the photographs of 3.2.

3.8.2 Markings.- All components of the training aid mockup shall be identified by use of numerals and titles and shall correspond to the nomenclature plate. The nomenclature plate shall be labeled as indicated on the photographs of 3.2. The title lettering shall be white on gray #26152.



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3.8.2 Markings.- All components of the training aid mockup shall be identified by use of numerals and titles and shall correspond to the nomenclature plate. The nomenclature plate shall be labeled as indicated on the photographs of 3.2.

3.8.2.1 Lettering.- All labels shall be of metal or plastic and permanently attached to the appropriate component. The information shall be engraved and the engraved line filled with a contrasting color. A non-reflective surface shall be used.

3.9 Workmanship.- The training aid, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of markings and indications of parts and assemblies, painting, and freedom of parts from burrs and sharp edges.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual and mechanical examination, and testing of materials, subassemblies, parts and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.4 Quality conformance inspection.- Quality conformance inspection shall consist of a preproduction inspection and inspection of production units.

4.4.1 Preproduction inspection.- A preproduction inspection shall be conducted on a preproduction unit and shall consist of all the examinations and tests specified in 4.5 and 4.6.



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4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction unit.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Design

(c) Construction

(d) Dimensions and tolerances

(e) Details of components

(f) Finish and color

(g) Nameplates and product markings

(h) Workmanship

(i) Preparation for delivery.

4.6 Tests.- Functional tests shall be conducted to determine compliance to the requirements of this specification.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified by the procuring activity.

5.1.1 Level A.- For Level A, each training aid shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.2 Level C.- For Level C, preservation and packaging shall conform to the manufacturer's commercial practice.

5.2 Packing.- Packing shall be Level A or C as specified by the procuring activity.

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2, or PPP-B-601, Styles A or B with steel banding. The gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547. A case liner shall not be required where the package(s) and package dunnage have been treated to resist water.



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5.2.2 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will ensure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation.

5.3 Physical protection.- Each training aid shall be packaged disassembled in a container, conforming to PPP-B-636, Class 2, designed to provide a compact and nonshifting load. Each assembly or part shall be individually wrapped in polyethylene or polyvinylidene chloride resin type of transparent film. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration, impact or abrasion during shipment and storage.

5.3.1 Cushioning materials used shall be compatible with all finishes so that migrating solvents in cements, cushioning or components will not affect any part of the training aid.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid mockup is intended to be used by instructors as an aid in group training of Naval personnel, or as a self-training medium in the theory of operation, construction, components, and the operation of a huddling chamber type safety valve.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification (see 1.1.1)

(b) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N252



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